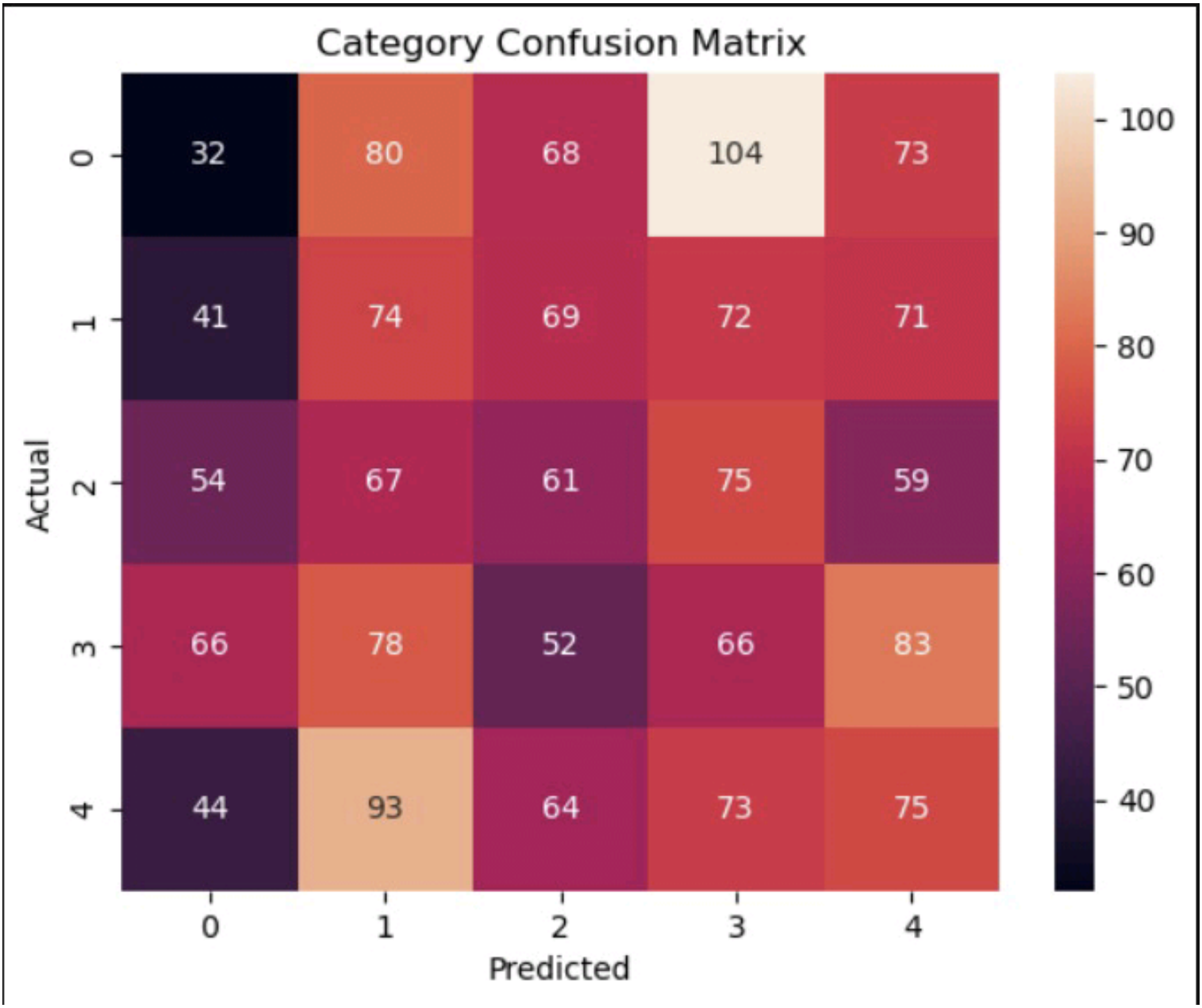
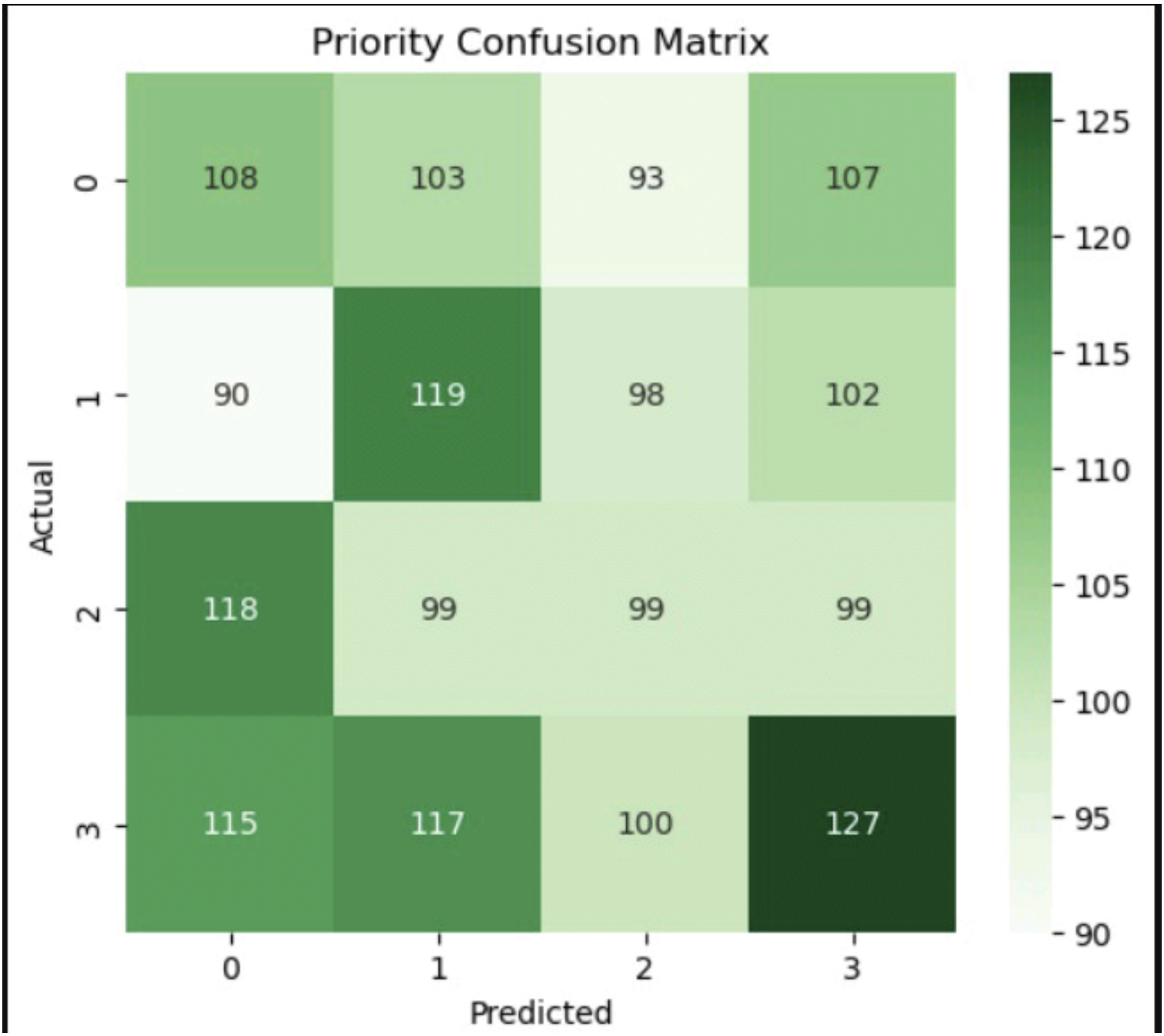
```
Category Model Accuracy: 0.18181818181818182

Category Classification Report:
                       precision    recall  f1-score   support

     Billing inquiry       0.14      0.09      0.11       357
Cancellation request       0.19      0.23      0.21       327
     Product inquiry       0.19      0.19      0.19       316
      Refund request       0.17      0.19      0.18       345
     Technical issue       0.21      0.21      0.21       349

            accuracy                           0.18      1694
           macro avg       0.18      0.18      0.18      1694
        weighted avg       0.18      0.18      0.18      1694

Priority Model Accuracy: 0.26741440377804016

Priority Classification Report:
               precision    recall  f1-score   support

    Critical       0.25      0.26      0.26       411
        High       0.27      0.29      0.28       409
         Low       0.25      0.24      0.25       415
      Medium       0.29      0.28      0.28       459

    accuracy                           0.27      1694
   macro avg       0.27      0.27      0.27      1694
weighted avg       0.27      0.27      0.27      1694
```


Category Confusion Matrix
Actual
Predicted
0
1
2
3
4
32
80
68
104
73
41
74
69
72
71
54
67
61
75
59
66
78
52
66
83
44
93
64
73
75
100
90
80
70
60
50
40


Priority Confusion Matrix
108
103
93
107
90
119
98
102
118
99
99
99
115
117
100
127
0
1
2
3
Actual
0
1
2
3
Predicted
125
120
115
110
105
100
95
90

[32]:

	Ticket ID	Customer Name	Customer Email	Customer Age	Customer Gender	Product Purchased	Date of Purchase	Ticket Type	Ticket Subject	Ticket Description	Ticket Status	Resolution	Ticket Priority
0	1	Marisa Obrien	carrollallison@example.com	32	Other	GoPro Hero	2021-03-22	Technical issue	Product setup	I'm having an issue with the {product_purchase...	Pending Customer Response	NaN	Critical
1	2	Jessica Rios	clarkeashley@example.com	42	Female	LG Smart TV	2021-05-22	Technical issue	Peripheral compatibility	I'm having an issue with the {product_purchase...	Pending Customer Response	NaN	Critical
2	3	Christopher Robbins	gonzalestracy@example.com	48	Other	Dell XPS	2020-07-14	Technical issue	Network problem	I'm facing a problem with my {product_purchase...	Closed	Case maybe show recently my computer follow.	Low
3	4	Christina Dillon	bradleyolson@example.org	27	Female	Microsoft Office	2020-11-13	Billing inquiry	Account access	I'm having an issue with the {product_purchase...	Closed	Try capital clearly never color toward story.	Low
4	5	Alexander Carroll	bradleymark@example.com	67	Female	Autodesk AutoCAD	2020-02-04	Billing inquiry	Data loss	I'm having an issue with the {product_purchase...	Closed	West decision evidence bit.	Low

[33]:

	Ticket Description	clean_text
0	I'm having an issue with the {product_purchase...	im having an issue with the productpurchased p...
1	I'm having an issue with the {product_purchase...	im having an issue with the productpurchased p...
2	I'm facing a problem with my {product_purchase...	im facing a problem with my productpurchased t...
3	I'm having an issue with the {product_purchase...	im having an issue with the productpurchased p...
4	I'm having an issue with the {product_purchase...	im having an issue with the productpurchased p...